

- 1 A golf ball is hit from point A on the ground and moves through the air to point B. The path is illustrated in Fig. 1.1.

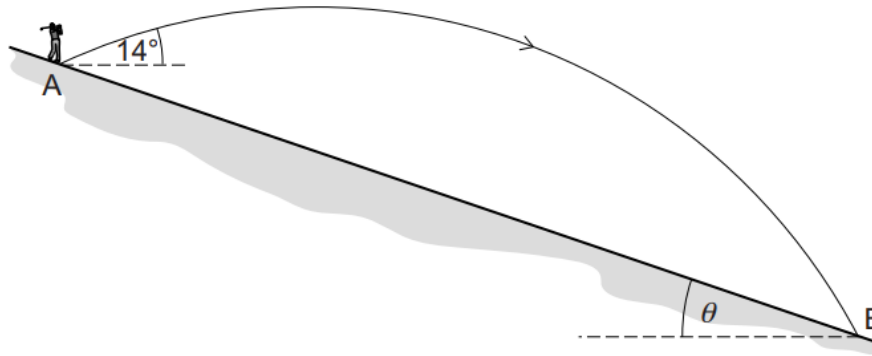


Fig. 1.1

The ground slopes downhill with a constant gradient. The ball has an initial speed of 67 m s^{-1} at an angle of 14° to the horizontal. The ball hits the ground at B after 4.9 s.

(a) Ignoring air resistance, calculate

- (i)** the horizontal and vertical components of the ball's velocity at A,

horizontal component at A = m s^{-1} [1]

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- (ii)** the horizontal displacement from A to B,

horizontal displacement = m [1]

(iii) the vertical displacement from A to B,

vertical displacement = m [2]

(iv) the angle of the slope to the horizontal, θ .

angle θ = ° [2]

(b) In a real situation, air resistance provides a force on the ball in the opposite direction to its motion.

(i) On Fig. 1.1, sketch a likely path of the ball hit from A when air resistance is taken into account. [1]

(ii) Give reasons for the shape you have drawn in **(b)(i)** for

1. the path of the ball at the start,

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 [1]

2. the angle at which the ball hits the ground.

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 [1]

2 (a) A satellite of mass 500 kg moves in a circular orbit at constant speed around the Earth